



Cognitive Biases and Logical Fallacies in Medical Practice

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AY (2025-2026)

Introduction

Cognitive bias in medicine and logical fallacies.

- These two concepts affect how we think, how we make decisions, and how we treat patients.
- Even the best doctors can fall into these traps.
- By the end of this lecture, you should be able to:
 1. Recognize common types of bias.
 2. Understand why they happen.
 3. Spot common logical fallacies.
 4. Avoid them when diagnosing patients or reading scientific information.

What is Bias?



Definition

- **Bias** means a “**systematic error in thinking.**”
It is not intentional.
It simply means our mind takes shortcuts, or relies on previous experiences, emotions, or expectations.
- In medicine, bias can lead to **wrong diagnosis, unsafe decisions, or poor patient communication.**

Common Types of Bias in Medicine



1. Anchoring Bias

- **Definition:**

Fixing your mind on the **first** piece of information and refusing to change your initial impression.

- **Example:**

A patient arrives with chest pain.

The nurse says, “It is probably gastric.”

The doctor immediately accepts this, and stops thinking about myocardial infarction or other causes.

2. Confirmation Bias

- **Definition:**

Looking **only** for **information** that **supports** what you already believe, and ignoring information that disagrees.

- **Example:**

A doctor thinks a child has a viral infection.

The child has fever and runny nose.

Even when the parent says, “She also has *neck stiffness*,” the doctor ignores it because it doesn’t fit his initial idea.

A serious condition like meningitis may be missed.

3. Availability Bias

- **Definition:**

Judging a condition as more likely because you recently saw a similar case.

- **Example:**

After seeing one case of influenza A, a doctor starts diagnosing all fever cases as influenza, even when symptoms do not match.

4. Overconfidence Bias

- **Definition:**

Believing too strongly in your own knowledge or experience.

- **Example:**

A senior doctor refuses to order a CT scan because he is “sure” it is not appendicitis.

But the patient actually has appendicitis.

5. Stereotyping Bias

- **Definition:**

Assuming something about a patient based on group identity (age, gender, culture, lifestyle).

- **Example:**

A young woman comes with chest pain.

The doctor assumes she is “too young” for a heart problem.

But young patients can still have serious cardiac issues.

Why Bias Happens ?

- The brain tries to save time.
- Doctors work under pressure and time limitations.
- Experience sometimes creates “mental shortcuts.”
- Fatigue, stress, and emotions affect judgment.

Recognizing bias is the first step to avoiding it.

LOGICAL FALLACIES IN MEDICINE

- **Definition**

A **logical fallacy** is an error in reasoning that makes an argument invalid.

- Patients use them.
- Students use them.
- Doctors use them too—often without noticing.

Understanding them helps you:

- Think clearly
- Make scientific decisions
- Evaluate medical research
- Counsel patients effectively

Common Logical Fallacies

1. Post Hoc Fallacy

- (*“After this, therefore because of this”*)
- **Definition:**
Assuming that because B happened after A, then A caused B.
- **Example:**
A patient says:
“I ate garlic and my blood pressure became normal. Garlic cured my hypertension.”
- But maybe the BP dropped naturally, or the patient was also taking medication.

2. Appeal to Authority

- (Because an important person said it, it must be true)
- **Example:**
“A famous doctor on TV recommends this supplement, so it must be effective.”
- ***Authority is not evidence.***
Only scientific studies are evidence.

3. Strawman Fallacy

- **Definition:**
Misrepresenting someone's argument to make it easier to attack
- **Example:**
Doctor says:
"We should encourage physical activity for mild hypertension."
Patient replies:
"So you are saying medicine is useless?"
- **This is a distortion of the original advice.**

4. Hasty Generalization

- **Definition:**
Making a conclusion from too little information.
- **Example:**
“I treated one patient with vitamin D and her bone pain improved.
Therefore, vitamin D cures all bone pain.”
- One case does not represent all cases.

Why Logical Fallacies Matter in Medicine?

- They mislead patients.
- They weaken scientific understanding.
- They cause incorrect clinical decisions.
- They fuel misinformation (especially online and in social media).

As medical students, you must learn to think critically, not emotionally.

How to Avoid Bias & Fallacies

1. Always consider multiple diagnoses (the differential diagnosis).

Don't stop at your first idea.

2. Ask for evidence.

“What study supports this?” “How many patients were included?”

3. Slow down when making decisions.

Even a 30-second pause improves accuracy.

4. Work in teams.

Listen to nurses, residents, colleagues. They may notice what you missed.

5. Challenge your own thinking.

Ask yourself: “What if I am wrong?” “What other explanation fits the symptoms?”

6. Learn from feedback.

Mistakes are teachers if you are willing to examine them honestly.

Conclusion

- Bias and logical fallacies are part of human thinking. But being aware of them will make you a safer, smarter, and more scientific doctor.
- Medicine is not only about anatomy and physiology— It is also about **clear thinking, good judgment, and open-minded**